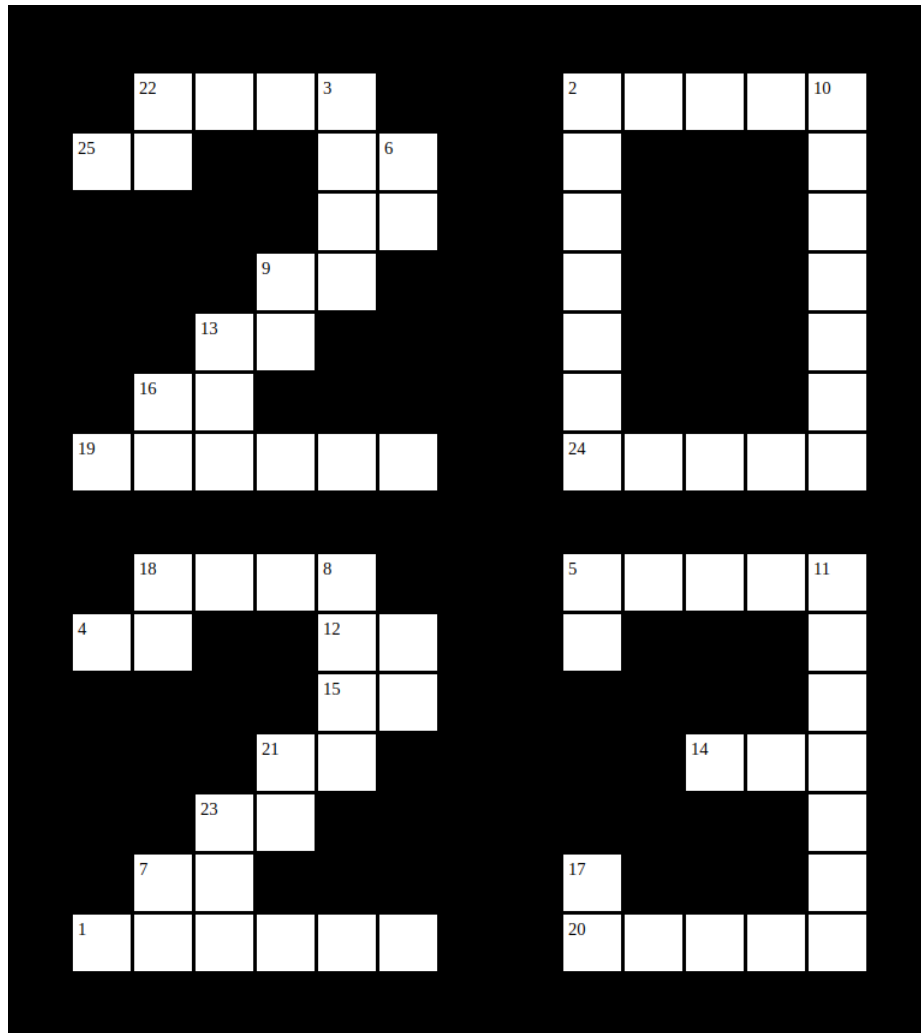


Number Crossword

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1 Puzzle



1.1 Introduction + Rules

This crossword contains clues that have been written by various Math Survey members and friends who have been kind enough to contribute to this puzzle. Their names are credited next to the clues that they have written, thank you so much to everyone who submitted a clue!

Rules: This puzzle works like a standard crossword, only the answers are numbers. Answer the clues for across and down to fill in the puzzle!

Across

1. $(2023 \div 17) - 90)(2023 + 2628)$
2. $(2023 - 2020)(2023 + 4120)$
4. Let S be the sum of the digits of 33^{22} . Given that:
 - The first 10 digits of 33^{22} are 2, 5, 5, 4, 5, 0, 4, 5, 3, 0 respectively,
 - The 27th through 33rd digits are 0, 4, 9, 5, 1, 4, 8 respectively,
 - Digits 17 and 18 are 7 and 4,
 - The average of all the remaining even-placed digits except for the last is an integer,
 - The difference of the remaining odd-placed digits and the remaining even-placed digits except for the last is a multiple of 5,

Find the product of $\frac{S-1}{8}$ and the smallest factor of S other than 1.
Credit: Patrick Was
5. $(2023 \times 36) + 1195$
7. This number is the sum of the first 6 triangular numbers (numbers where given a triangular number of objects, you can arrange those objects into an equilateral triangle).
12. $\int_{-12}^{12} \sqrt[3]{x - \lfloor x \rfloor} dx = ?$
Credit: Zawad Dewan
14. This number is a three digit palindromic number equal to the product of the sum of its digits and ten more than the product of its digits.
15. Compute the number of factors of $196^2 + 578^2$.
Credit: Julia Kozak
18. $((2023 \div 119) + 20) \times 77$
19. $(2023 \times 243) + 716$
20. $((2023 + 75) \times 40$
21. Let n be the number of ordered triples of integers (x,y,z) are there such that $x^2 + y^2 + z^2 = 34$. Compute $2n - 5$.
Credit: Kathleen Zhang
22. $(2023 - 1940) \times 42$
23. $(2023 \div 17) - 45$
24. Rin is deciding how to auction a piece of jewelry to Elie and Josiah. Her first option is to limit both bidders to one bid hidden from each other, and have the winner pay the loser's bid for the jewelry. Her second option is to let each bidder bid multiple times, but force both bidders to pay their own final bids, even though only the winner will get the jewelry. Elie and Josiah value the jewelry at \$127 and \$151 respectively, and both bring \$10 more than they value their jewelry at. Both bidders decide to submit bids, bid optimally in integer amounts

and don't communicate with each other. If Elie bids first, what is the product of Josiah's final bids in each of Rin's potential auction plans?

Credit: Lee Eisenberg

25. $(2023 \div 119) + 2$

Down

2. $((2023 \times 375) + 106) \times 2$

3. $(2023 \times 3) + 245$

5. Find

$$\lim_{n \rightarrow \infty} \frac{\sum_{k=1}^n (.99)^k}{\sum_{k=1}^n \left(\frac{2k+2}{k^2+2k} - \frac{2k+4}{k^2+4k+3} \right)}$$

Credit: Ethan Sharma

6. $(2023 \div 119) + 70$

8. $(2023 \times 4) + (20230 \div 17) - 151$

9. Find the largest prime under 100 that is the sum of another prime and a positive integer to the fifth power.

Credit: Daphne Qin

10. $(4046 \times 2023) + (2023 \times 23^2) + (2023 \times 23 \times 6) + (2023 \times 19) + 82$

11. $2023^2 + -(2023 \times 460) + 1871$

13. $2023 - 1761$

16. $(2023 \div 17) - 60$

17. Let the sequence N be

$$\frac{2}{2 \times 3} + \frac{2}{3 \times 8} + \frac{2}{4 \times 15} + \frac{2}{5 \times 24} + \dots$$

. The sum of the first n th terms of the sequence N is $\frac{189}{390}$. Compute the value of n .

Credit: Lucia Liu

18. Find the smallest two-digit number that is divisible by the number of primes less than the number.

Credit: Mr. Kats

The solution is on page 6.